

CHAPTER 16

PERMUTATIONS

16.1 THE FACTORIAL

In this section, we shall introduce the term and notation of factorial which will be often used in this chapter and the next three chapters.

FACTORIAL The continued product of first n natural numbers is called the " n factorial" and is denoted by $n!$ or $\underline{n}$.

i.e. $n! = 1 \times 2 \times 3 \times 4 \times \dots \times (n-1) \times n$.

Thus, $3! = 1 \times 2 \times 3 = 6$; $4! = 1 \times 2 \times 3 \times 4 = 24$, $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$ etc.

Clearly, $n!$ is defined for positive integers only.

ZERO FACTORIAL As we will require zero factorial in the later sections of this chapter and it does not make any sense to define it as the product of the integers from 1 to zero. So, we define $0! = 1$.

NOTE Factorials of proper fractions or negative integers are not defined. Factorial n is defined only for whole numbers.

DEDUCTION We have,

$$n! = 1 \times 2 \times 3 \times 4 \dots \times (n-1) \times n = [1 \times 2 \times 3 \times 4 \dots \times (n-1)] n = [(n-1)!] n = n \times (n-1)!$$

Thus, $n! = n \times (n-1)!$

Similarly,

$$n! = n(n-1)(n-2)! = n(n-1)(n-2)(n-3)! = n(n-1)(n-2)(n-3)(n-4)! \text{ and so on.}$$

For example, $8! = 8(7!)$, $5! = 5(4!)$ and $2! = 2(1!)$

Following examples will illustrate the use of this property of factorial n .

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 ✓ Compute: (i) $\frac{20!}{18!}$ (ii) $\frac{10!}{6!4!}$

SOLUTION (i) We have,

$$\begin{aligned} \frac{20!}{18!} &= \frac{20(19!)}{18!} = \frac{20 \times 19 \times 18!}{18!} \\ &= 20 \times 19 = 380 \end{aligned}$$

$$[\because n! = n \times (n-1)!]$$

$$(ii) \quad \frac{10!}{6!4!} = \frac{10 \times 9 \times 8 \times 7 \times 6!}{6! \times (4 \times 3 \times 2 \times 1)} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

EXAMPLE 2 ✓ Convert the following products into factorials:

(i) $6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$

(ii) $2 \cdot 4 \cdot 6 \cdot 8 \cdot 10$

SOLUTION (i) $6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = \frac{10!}{5!}$

(ii) $2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 = (2 \times 1)(2 \times 2)(2 \times 3)(2 \times 4)(2 \times 5) = 2^5 \times (1 \cdot 2 \cdot 3 \cdot 4 \cdot 5) = 2^5 \times 5!$

EXAMPLE 3 Find the LCM of $4!$, $5!$ and $6!$

SOLUTION We have, $5! = 5 \times 4!$ and $6! = 6 \times 5 \times 4!$

$\therefore$ L.C.M. of $4!$, $5!$, $6! = \text{L.C.M. } \{4!, 5 \times 4!, 6 \times 5 \times 4!\} = (4!) \times 5 \times 6 = 6! = 720$

EXAMPLE 4 If $\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$, find x .

SOLUTION We have,

$$\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$$

$$\Rightarrow \frac{1}{9!} + \frac{1}{10 \times 9!} = \frac{x}{11 \times 10 \times 9!}$$

$$\Rightarrow \frac{1}{9!} \left(1 + \frac{1}{10} \right) = \left(\frac{x}{11 \times 10} \right) \times \frac{1}{9!}$$

$$\Rightarrow 1 + \frac{1}{10} = \frac{x}{11 \times 10} \Rightarrow \frac{11}{10} = \frac{x}{11 \times 10} \Rightarrow x = 11 \times 11 = 121.$$

ALITER We have,

$$\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$$

Multiplying both sides by the LCM of $9!$, $10!$ and $11!$ i.e. by $11!$, we obtain

$$\frac{11!}{9!} + \frac{11!}{10!} = \frac{x}{11!} \times 11!$$

$$\Rightarrow \frac{11 \times 10 \times 9!}{9!} + \frac{11 \times 10!}{10!} = x$$

$$\Rightarrow 11 \times 10 + 11 = x$$

$$\Rightarrow x = 121.$$

EXAMPLE 5 Find n , if:

(i) $(n+2)! = 2550 \times n!$

(ii) $(n+1)! = 12 \times (n-1)!$

SOLUTION (i) We have,

$$(n+2)! = 2550 \times n!$$

$$\Rightarrow (n+2)(n+1) \times n! = 2550 \times n!$$

$$\Rightarrow (n+2) \times (n+1) = 2550$$

$$\Rightarrow (n+2)(n+1) = 51 \times 50$$

$$\Rightarrow n+2 = 51 \text{ or } n+1 = 50$$

$$\Rightarrow n = 49$$

(ii) We have,

$$(n+1)! = 12 \times (n-1)!$$

$$\Rightarrow (n+1) \times n \times (n-1)! = 12 \times (n-1)!$$

$$\Rightarrow n(n+1) = 12$$

$$\Rightarrow (n+1)n = 4 \times 3 \Rightarrow n = 3$$

[Expressing 2550 as the product of two consecutive natural numbers]

[By comparing]

[By comparing]

EXAMPLE 6 If $\frac{n!}{2!(n-2)!}$ and $\frac{n!}{4!(n-4)!}$ are in the ratio 2 : 1, find the value of n .

SOLUTION We have,

$$\frac{n!}{2!(n-2)!} : \frac{n!}{4!(n-4)!} = 2 : 1$$

$$\Rightarrow \frac{n!}{2!(n-2)!} \times \frac{4!(n-4)!}{n!} = \frac{2}{1}$$

$$\Rightarrow \frac{4!(n-4)!}{2!(n-2) \times (n-3) \times (n-4)!} = \frac{2}{1}$$

$$\Rightarrow \frac{4 \times 3 \times 2!}{2!(n-2)(n-3)} = \frac{2}{1}$$

$$\Rightarrow (n-2)(n-3) = 6 \Rightarrow (n-2)(n-3) = 3 \times 2 \Rightarrow n-2 = 3 \text{ and } n-3 = 2 \Rightarrow n = 5$$

EXAMPLE 7 Prove that: $\frac{(2n)!}{n!} = \{1 \cdot 3 \cdot 5 \dots (2n-1)\} 2^n$.

SOLUTION We have,

$$\begin{aligned} \frac{(2n)!}{n!} &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \dots (2n-2)(2n-1)(2n)}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} \cdot \{2 \cdot 4 \cdot 6 \cdot 8 \dots (2n-2)(2n)\}}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} 2^n \{1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1)n\}}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} \cdot 2^n \cdot n!}{n!} = \{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} 2^n \end{aligned}$$

LEVEL-2

EXAMPLE 8 Prove that $(n! + 1)$ is not divisible by any natural number between 2 and n .

SOLUTION Let m be divisible by k and r be any natural number between 1 and k . If $m + r$ is divided by k , then we obtain r as the remainder.

We have, $n! = 1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1) \cdot n$.

Therefore, $n!$ is divisible by every natural number between 2 and n . So, $(n! + 1)$, when divided by any natural number between 2 and n , leaves 1 as the remainder.

Hence, $(n! + 1)$ is not divisible by any natural number between 2 and n .

EXAMPLE 9 Prove the inequalities $(n!)^2 \leq n^n (n!) < (2n)!$ for all positive integers n .

SOLUTION Clearly,

$$(n!)^2 = (n!)(n!) = (1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1)n)(n!)$$

$$\text{Now, } \left. \begin{array}{l} 1 \leq n \\ 2 \leq n \\ 3 \leq n \\ \vdots \\ (n-1) \leq n \\ n \leq n \end{array} \right\} \Rightarrow 1 \cdot 2 \cdot (n-1)n \leq n \cdot n \cdot n \dots n \quad n\text{-times}$$

$$\Rightarrow n! \leq n^n \Rightarrow (n!)(n!) \leq n^n (n!) \Rightarrow (n!)^2 \leq n^n (n!) \quad \dots(i)$$

We have, $(2n)! = 1 \cdot 2 \cdot 3 \dots (n-1)n(n+1)(n+2) \dots (2n-1)(2n)$

$$\Rightarrow (2n)! = n!(n+1)(n+2)\dots(2n-1)(2n)$$

$$\text{Now, } \left. \begin{array}{l} n+1 > n \\ n+2 > n \\ n+3 > n \\ \vdots \\ n+(n-1) > n \\ n+n > n \end{array} \right\} \Rightarrow (n+1)(n+2)(n+3)\dots(2n-1)(2n) > n^n$$

$$\Rightarrow n!(n+1)(n+2)\dots(2n-1)(2n) > n!n^n$$

$$\Rightarrow (2n)! > n!n^n \Rightarrow n!n^n < (2n)! \quad \dots(\text{ii})$$

From (i) and (ii), we get $(n!)^2 \leq n^n (n!) < (2n)!$

EXAMPLE 10 Prove that $33!$ is divisible by 2^{15} . What is the largest integer n such that $33!$ is divisible by 2^n ?

SOLUTION Let $E_2(n)$ denote the index of 2 in n . Then,

$$E_2(33!) = E_2(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \dots 32 \cdot 33)$$

$$\Rightarrow E_2(33!) = E_2(2 \cdot 4 \cdot 6 \cdot 8 \dots 30 \cdot 32)$$

$$\Rightarrow E_2(33!) = 16 + E_2(1 \cdot 2 \cdot 3 \dots 15 \cdot 16)$$

$$\Rightarrow E_2(33!) = 16 + E_2(2 \cdot 4 \cdot 6 \dots 14 \cdot 16)$$

$$\Rightarrow E_2(33!) = 16 + 8 + E_2(1 \cdot 2 \cdot 3 \dots 8)$$

$$\Rightarrow E_2(33!) = 16 + 8 + E_2(2 \cdot 4 \cdot 6 \cdot 8)$$

$$\Rightarrow E_2(33!) = 16 + 8 + 4 + E_2(1 \cdot 2 \cdot 3 \cdot 4)$$

$$\Rightarrow E_2(33!) = 16 + 8 + 4 + E_2(2 \cdot 4) = 16 + 8 + 4 + 3 = 31.$$

Thus, exponent of 2 in $33!$ is 31 i.e. $33! = 2^{31} \times \text{an integer}$

This shows that $33!$ is divisible by 2^{15} and the largest integer n such that $33!$ is divisible by 2^n is 31.

EXERCISE 16.1

LEVEL-1

1. Compute:

(i) $\frac{30!}{28!}$

(ii) $\frac{11! - 10!}{9!}$

(iii) L.C.M. $(6!, 7!, 8!)$

2. Prove that $\frac{1}{9!} + \frac{1}{10!} + \frac{1}{11!} = \frac{122}{11!}$

3. Find x in each of the following:

(i) $\frac{1}{4!} + \frac{1}{5!} = \frac{x}{6!}$

(ii) $\frac{x}{10!} = \frac{1}{8!} + \frac{1}{9!}$

[NCERT]

(iii) $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$

[NCERT]

4. Convert the following products into factorials:

(i) $5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$

(ii) $3 \cdot 6 \cdot 9 \cdot 12 \cdot 15 \cdot 18$

(iii) $(n+1)(n+2)(n+3)\dots(2n)$

(iv) $1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \dots (2n-1)$

5. Which of the following are true:

(i) $(2+3)! = 2! + 3!$

(ii) $(2 \times 3)! = 2! \times 3!$

6. Prove that $n!(n+2) = n! + (n+1)!$

7. If $(n+2)! = 60[(n-1)!]$, find n .

8. If $(n+1)! = 90[(n-1)!]$, find n .

9. If $(n+3)! = 56[(n+1)!]$, find n .

10. If $\frac{(2n)!}{3!(2n-3)!}$ and $\frac{n!}{2!(n-2)!}$ are in the ratio 44 : 3, find n .

11. Prove that:

$$(i) \frac{n!}{(n-r)!} = n(n-1)(n-2) \dots (n-(r-1)) \quad (ii) \frac{n!}{(n-r)!r!} + \frac{n!}{(n-r+1)!(r-1)!} = \frac{(n+1)!}{r!(n-r+1)!}$$

12. Prove that: $\frac{(2n+1)!}{n!} = 2^n \{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)\}$

ANSWERS

1. (i) 870 (ii) 100 (iii) 8! 3. (i) 36 (ii) 100 (iii) 64 4. (i) $\frac{10!}{4!}$ (ii) $3^6(6!)$
 (iii) $\frac{(2n)!}{n!}$ (iv) $\frac{(2n)!}{2^n n!}$ 5. (i) False (ii) False 7. 3 8. 9 9. 5 10. 6

HINTS TO NCERT & SELECTED PROBLEMS

3. (i) We have,

$$\frac{1}{4!} + \frac{1}{5!} = \frac{x}{6!}$$

$$\Rightarrow \frac{6!}{4!} + \frac{6!}{5!} = x$$

[Multiplying both sides by 6!]

$$\Rightarrow \frac{6 \times 5 \times 4!}{4!} + \frac{6 \times 5!}{5!} = x \Rightarrow 6 \times 5 + 6 = x \Rightarrow x = 36$$

(ii) We have,

$$\frac{x}{10!} = \frac{1}{8!} + \frac{1}{9!}$$

$$\Rightarrow x = \frac{10!}{8!} + \frac{10!}{9!}$$

[Multiplying both sides by 10!]

$$\Rightarrow x = \frac{10 \times 9 \times 8!}{8!} + \frac{10 \times 9!}{9!}$$

$$\Rightarrow x = 10 \times 9 + 10 = 100$$

(iii) We have,

$$\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$$

$$\Rightarrow \frac{8!}{6!} + \frac{8!}{7!} = x$$

[Multiplying both sides by 8!]

$$\Rightarrow \frac{8 \times 7 \times 6!}{6!} + \frac{8 \times 7!}{7!} = x$$

$$\Rightarrow 8 \times 7 + 8 = x$$

$$\Rightarrow x = 64$$

7. $(n+2)! = 60(n-1)!$

$$\Rightarrow (n+2)(n+1)(n)(n-1)! = 60 \times (n-1)!$$

$$\Rightarrow (n+2)(n+1)(n) = 5 \times 4 \times 3 \quad [\text{Expressing 60 as the product of three consecutive integers}]$$

$$\Rightarrow n = 3$$

[On comparing two sides]

8. $(n+1)! = 90(n-1)!$

$$\Rightarrow n+1(n)(n-1)! = 90(n-1)!$$

$$\Rightarrow (n+1)n = 10 \times 9$$

[Writing 90 as the product of consecutive integers]

$$\Rightarrow n = 9$$

$$9. (n+3)! = 56(n+1)!$$

$$\Rightarrow (n+3)(n+2)(n+1)! = 56(n+1)!$$

$$\Rightarrow (n+3)(n+2) = 8 \times 7$$

[Writing 56 as the product of consecutive integers]

$$\Rightarrow n+2=7 \Rightarrow n=5$$

16.2 FUNDAMENTAL PRINCIPLES OF COUNTING

In this section, we shall discuss two fundamental principles viz. principle of addition and principle of multiplication. These two principles will enable us to understand permutations and combinations. In fact these two principles form the base of permutations and combinations.

FUNDAMENTAL PRINCIPLE OF MULTIPLICATION *If there are two jobs such that one of them can be completed in m ways, and when it has been completed in any one of these m ways, second job can be completed in n ways; then the two jobs in succession can be completed in $m \times n$ ways.*

EXPLANATION If the first job is performed in any one of the m ways, we can associate with this any one of the n ways of performing the second job; and thus there are n ways of performing the two jobs without considering more than one way of performing the first; and so corresponding to each of the m ways of performing the first job, we have n ways of performing the second job. Hence, the number of ways in which the two jobs can be performed is $m \times n$.

ILLUSTRATION 1 *In a class there are 10 boys and 8 girls. The teacher wants to select a boy and a girl to represent the class in a function. In how many ways can the teacher make this selection?*

SOLUTION Here the teacher is to perform two jobs:

- (i) selecting a boy among 10 boys, and (ii) selecting a girl among 8 girls.

The first of these can be performed in 10 ways and the second in 8 ways. Therefore by the fundamental principle of multiplication, the required number of ways is $10 \times 8 = 80$.

REMARK *The above principle can be extended for any finite number of jobs as stated below:*

If there are n jobs $J_1, J_2, \dots, J_n$ such that job J_i can be performed independently in m_i ways; $i = 1, 2, \dots, n$. Then the total number of ways in which all the jobs can be performed is $m_1 \times m_2 \times m_3 \times \dots \times m_n$.

FUNDAMENTAL PRINCIPLE OF ADDITION *If there are two jobs such that they can be performed independently in m and n ways respectively, then either of the two jobs can be performed in $(m + n)$ ways.*

ILLUSTRATION 2 *In a class there are 10 boys and 8 girls. The teacher wants to select either a boy or a girl to represent the class in a function. In how many ways the teacher can make this selection?*

SOLUTION Here the teacher is to perform either of the following two jobs:

- (i) selecting a boy among 10 boys. or, (ii) selecting a girl among 8 girls.

The first of these can be performed in 10 ways and the second in 8 ways. Therefore, by fundamental principle of addition either of the two jobs can be performed in $(10 + 8) = 18$ ways. Hence, the teacher can make the selection of either a boy or a girl in 18 ways.

DIFFERENCE BETWEEN THE TWO PRINCIPLES *As we have discussed in the principle of multiplication a job is divided or decomposed into a number of sub-jobs which are unconnected to each other and the job is said to be performed if each sub-job is performed. While in the principle of addition there are a number of independent jobs and we have to perform one of them. So, the total number of ways of completing any one of the sub-jobs is the sum of the number of ways of completing each sub-jobs.*

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 *There are 3 candidates for a Classical, 5 for a Mathematical, and 4 for a Natural science scholarship.*

- (i) *In how many ways can these scholarships be awarded?*
 (ii) *In how many ways one of these scholarships be awarded?*

SOLUTION Clearly, Classical scholarship can be awarded to any one of the three candidates. So, there are 3 ways of awarding the Classical scholarship.

Similarly, Mathematical and Natural science scholarships can be awarded in 5 and 4 ways respectively. So, by Fundamental Principle of multiplication,

$$\text{Number of ways of awarding three scholarships} = 3 \times 5 \times 4 = 60$$

By Fundamental Principle of addition,

$$\text{Number of way of awarding one of the three scholarships} = 3 + 5 + 4 = 12$$

EXAMPLE 2 A room has 6 doors. In how many ways can a man enter the room through one door and come out through a different door ?

SOLUTION Clearly, a person can enter the room through any one of the six doors. So, there are six ways of entering into the room. After entering into the room, the man can come out through any one of the remaining five doors. So, he can come out through a different door in 5 ways.

Hence, the number of ways in which a man can enter a room through one door and come out through a different door = $6 \times 5 = 30$.

EXAMPLE 3 The flag of a newly formed forum is in the form $\square\square\square$ of three blocks, each to be coloured differently. If there are six different colours on the whole to choose from, how many such designs are possible ?

SOLUTION Since there are six colours to choose from, therefore, first block can be coloured in 6 ways. Now, the second block can be coloured by any one of the remaining colours in five ways. So, there are five ways to colour the second block.

After colouring first two blocks only four colours are left. The third block can now be coloured by any one of the remaining four colours. So, there are four ways to colour the third block.

Hence, by the fundamental principle of multiplication, the number of flag-designs is $6 \times 5 \times 4 = 120$.

EXAMPLE 4 Find the number of 4 letter words, with or without meaning, which can be formed out of the letters of the word ROSE, when [NCERT]

- (i) the repetition of the letters is not allowed. (ii) the repetition of the letters is allowed.

SOLUTION (i) The total number of words is same as the number of ways of filling in 4 vacant places $\square\square\square\square$ by the 4 letters. The first place can be filled in 4 different ways by any one of the 4 letters R, O, S, E. Since the repetition of letters is not allowed. Therefore, the second place can be filled in by any one of the remaining 3 letters in 3 different ways, following which the third place can be filled in by the remaining 2 letters in 2 different ways; following which the fourth place can be filled in by the remaining one letter in one way. Thus, by the fundamental principle of counting the required number of ways is $4 \times 3 \times 2 \times 1 = 24$.

Hence, required number of words = 24.

(ii) If the repetition of the letters is allowed, then each of the 4 vacant places can be filled in succession in 4 different ways.

Hence, required number of words = $4 \times 4 \times 4 \times 4 = 256$.

EXAMPLE 5 Given 4 flags of different colours, how many different signals can be generated, if a signal requires the use of 2 flags one below the other? [NCERT]

SOLUTION The total number of signals is equal to the number of ways of filling in 2 vacant places $\square\square$ in succession by four flags of different colours. The upper vacant place can be filled in 4 different ways by any one of the 4 flags; following which, the lower vacant place can be filled in 3 different ways by any one of the remaining the different flags.

Hence, by the fundamental principle of multiplication, the required number of signals is $4 \times 3 = 12$.

EXAMPLE 6 Find the number of different signals that can be generated by arranging at least 2 flags in order (one below the other) on a vertical staff, if five different flags are available. [NCERT]

SOLUTION Since a signal may consist of either 2 flags, 3 flags, 4 flags or 5 flags. Therefore,

$$\begin{aligned}
 \text{Total number of signals} &= \text{Number of 2 flags signals} \quad \text{=====} \\
 &+ \text{Number of 3 flags signals} \quad \text{=====} \\
 &+ \text{Number of 4 flags signals} \quad \text{=====} \\
 &+ \text{Number of 5 flags signals} \quad \text{=====} \\
 &= 5 \times 4 + 5 \times 4 \times 3 + 5 \times 4 \times 3 \times 2 + 5 \times 4 \times 3 \times 2 \times 1 \\
 &= 20 + 60 + 120 + 120 = 320
 \end{aligned}$$

EXAMPLE 7 In a monthly test, the teacher decides that there will be three questions, one from each of Exercises 7, 8 and 9 of the text book. If there are 12 questions in Exercise 7, 18 in Exercise 8 and 9 in Exercise 9, in how many ways can three questions be selected?

SOLUTION There are 12 questions in exercise 7. So, one question from exercise 7 can be selected in 12 ways. Exercise 8 contains 18 questions. So, second question can be selected in 18 ways. There are 9 questions in exercise 9. So, third question can be selected in 9 ways. Hence, three questions can be selected in $12 \times 18 \times 9 = 1944$ ways.

EXAMPLE 8 How many words (with or without meaning) of three distinct letters of the English alphabets are there?

SOLUTION Here we have to fill up three places by distinct letters of the English alphabets. Since there are 26 letters of the English alphabet, the first place can be filled by any of these letters. So, there are 26 ways of filling up the first place. Now, the second place can be filled up by any of the remaining 25 letters. So, there are 25 ways of filling up the second place. After filling up the first two places only 24 letters are left to fill up the third place. So, the third place can be filled in 24 ways.

Hence, the required number of words $= 26 \times 25 \times 24 = 15600$

EXAMPLE 9 There are 6 multiple choice questions in an examination. How many sequence of answers are possible, if the first three questions have 4 choices each and the next three have 5 each?

SOLUTION Here we have to perform 6 jobs of answering 6 multiple choice questions. Each one of the first three questions can be answered in 4 ways and each one of the next three can be answered in 5 different ways.

So, the total number of different sequences $= 4 \times 4 \times 4 \times 5 \times 5 \times 5 = 8000$

EXAMPLE 10 Find the total number of ways of answering 5 objective type questions, each question having 4 choices.

SOLUTION Since each question can be answered in 4 ways. So, the total number of ways of answering 5 questions is $4 \times 4 \times 4 \times 4 \times 4 = 4^5$.

EXAMPLE 11 How many three-digit numbers can be formed without using the digits 0, 2, 3, 4, 5 and 6?

SOLUTION We have to determine the total number of three digit numbers formed by using the digits 1, 7, 8, 9. Clearly, the repetition of digits is allowed.

A three digit number has three places viz. units's, ten's and hundred's. Unit's place can be filled by any of the digits 1, 7, 8, 9. So, unit's place can be filled in 4 ways. Similarly, each one of the ten's and hundred's place can be filled in 4 ways.

$\therefore$ Total number of required numbers $= 4 \times 4 \times 4 = 64$.

EXAMPLE 12 How many numbers are there between 100 and 1000 in which all the digits are distinct?

SOLUTION A number between 100 and 1000 has three digits. So, we have to form all possible 3-digit numbers with distinct digits. We cannot have 0 at the hundred's place. So, the hundred's place can be filled with any of the 9 digits 1, 2, 3, ..., 9. So, there are 9 ways of filling the hundred's place.

Now, 9 digits are left including 0. So, ten's place can be filled with any of the remaining 9 digits in 9 ways. Now, the unit's place can be filled with in any of the remaining 8 digits. So, there are 8 ways of filling the unit's place.

Hence, the total number of required numbers $= 9 \times 9 \times 8 = 648$.

EXAMPLE 13 *How many numbers are there between 100 and 1000 such that every digit is either 2 or 9?*

SOLUTION Every number between 100 and 1000 consists of three digits. So, we have to determine the total number of three digit numbers such that every digit is either 2 or 9.

Clearly, each one of the unit's, ten's and hundred's place can be filled in 2 ways.

So, the total number of required numbers $= 2 \times 2 \times 2 = 8$.

EXAMPLE 14 *How many numbers are there between 100 and 1000 such that 7 is in the unit's place.*

SOLUTION Every number between 100 and 1000 is a three digit number. So, we have to form 3-digit numbers with 7 at the unit's place by using the digits 0, 1, 2, ..., 9. Clearly, repetition of digits is allowed. The hundred's place can be filled with any of the digits from 1 to 9 (zero cannot be there at hundred's place). So, hundred's place can be filled in 9 ways. Now, the ten's place can be filled with any of the digits from 0 to 9. So, ten's place can be filled in 10 ways. Since all the numbers have digit 7 at the unit's place, so, unit's place can be filled in only one way. Hence, by the fundamental principle of counting the total number of numbers between 100 and 1000 having 7 at the unit's place $= 9 \times 10 \times 1 = 90$.

EXAMPLE 15 *A gentleman has 6 friends to invite. In how many ways can he send invitation cards to them, if he has three servants to carry the cards?*

SOLUTION Since a card can be sent by any one of the three servants, so the number of ways of sending the invitation card to the first friend $= 3$. Similarly, invitation cards can be sent to each of the six friends in 3 ways.

So, the required number of ways $= 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^6 = 729$.

EXAMPLE 16 *How many three-digit numbers more than 600 can be formed by using the digits 2, 3, 4, 6, 7.*

SOLUTION Clearly, repetition of digits is allowed. Since a three-digit number greater than 600 will have 6 or 7 at hundred's place. So, hundred's place can be filled in 2 ways. Each of the ten's and one's place can be filled in 5 ways.

Hence, total number of required numbers $= 2 \times 5 \times 5 = 50$.

EXAMPLE 17 *How many numbers between 3000 and 4000 can be formed from the digits 3, 4, 5, 6, 7 and 8, no digit being repeated in any number?*

SOLUTION Clearly, a number between 3000 and 4000 must have 3 at thousand's place. So, thousand's place can be filled in only one way. Now, hundred's place can be filled in 5 ways. Since repetition of digits is not allowed so ten's and one's places can be filled in 4 and 3 ways respectively.

So, total number of required numbers $= 1 \times 5 \times 4 \times 3 = 60$.

EXAMPLE 18 *How many numbers divisible by 5 and lying between 4000 and 5000 can be formed from the digits 4, 5, 6, 7 and 8.*

SOLUTION Clearly, a number between 4000 and 5000 must have 4 at thousand's place. Since the number is divisible by 5 it must have 5 at unit's place. Now, each of the remaining places (viz. hundred's and ten's) can be filled in 5 ways.

Hence, total number of required numbers $= 1 \times 5 \times 5 \times 1 = 25$.

EXAMPLE 19 How many four-digit numbers can be formed using the digits 0, 1, 2, 3, 4, 5 if (i) repetition of digits is not allowed (ii) repetition of digits is allowed ?

SOLUTION (i) In a four-digit number 0 cannot appear in the thousand's place. So, thousand's place can be filled in 5 ways. (viz. 1, 2, 3, 4, 5). Since repetition of digits is not allowed and 0 can be used at hundred's place, so hundred's place can be filled in 5 ways.

Now, any one of the remaining four digits can be used to fill up ten's place. So, ten's place can be filled in 4 ways. One's place can be filled from the remaining three digits in 3 ways.

Hence, the required number of numbers $= 5 \times 5 \times 4 \times 3 = 300$.

(ii) For a four-digit number we have to fill up four places and 0 cannot appear in the thousand's place. So, thousand's place can be filled in 5 ways. Since repetition of digits is allowed, so each of the remaining three places viz. hundred's, ten's and one's can be filled in 6 ways.

Hence, the required number of numbers $= 5 \times 6 \times 6 \times 6 = 1080$.

EXAMPLE 20 How many numbers greater than 1000, but not greater than 4000 can be formed with the digits 0, 1, 2, 3, 4 if : (i) repetition of digits is allowed ? (ii) repetition of digits is not allowed ?

SOLUTION (i) Every number between 1000 and 4000 is a four digit number. In thousand's place we can put either 1 or 2 or 3 but not 4. So, thousand's place can be filled in 3 ways. Since repetition of digits is allowed, so each of the hundred's, ten's and one's place can be filled in 5 ways. So, total number of numbers between 1000 and 4000, including 1000 and excluding 4000 is $3 \times 5 \times 5 \times 5 = 375$. But, we have to find the total number of numbers greater than 1000 but not greater than 4000.

Hence, required number of numbers $= 375 + 1 \text{ (for 4000)} - 1 \text{ (for 1000)} = 375$.

(ii) As discussed above thousand's place can be filled in 3 ways. Since repetition of digits is not allowed, so, hundred's place can be filled from the remaining digits in 4 ways. Now, three digits are left, so ten's place can be filled in 3 ways. One's place can be filled in 2 ways.

Hence, required number of numbers $= 3 \times 4 \times 3 \times 2 = 72$.

EXAMPLE 21 How many three digit odd numbers can be formed by using the digits 1, 2, 3, 4, 5, 6 if:

[NCERT]

(i) the repetition of digits is not allowed ? (ii) the repetition of digits is allowed ?

SOLUTION For a number to be odd, we must have 1, 3 or 5 at the unit's place. So, there are 3 ways of filling the unit's place.

(i) Since the repetition of digits is not allowed, the ten's place can be filled with any of the remaining 5 digits in 5 ways. Now, four digits are left. So, hundred's place can be filled in 4 ways.

So, required number of numbers $= 3 \times 5 \times 4 = 60$

(ii) Since the repetition of digits is allowed, so each of the ten's and hundred's place can be filled in 6 ways.

Hence, required number of numbers $= 3 \times 6 \times 6 = 108$.

EXAMPLE 22 How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated? [NCERT]

SOLUTION For a number to be even, we must have 2, 4 or 6 at the unit's place. So, there are 3 ways to fill in the unit's place. Since digits can be repeated, so each of the ten's and hundred's place can be filled in 6 ways.

EXAMPLE 23 How many numbers of 3 digits can be formed with the digits 1, 2, 3, 4, 5 when digits may be repeated ?

SOLUTION The unit's place can be filled in 5 ways. Since, the repetition of digits is allowed, therefore ten's place can be filled in 5 ways and hundred's place can also be filled in 5 ways.

Therefore, by the fundamental principle of counting, the required number of three digit numbers $= 5 \times 5 \times 5 = 125$.

EXAMPLE 24 Find the number of numbers of 5 digits that can be formed with the digits 0, 1, 2, 3, 4 if the digits can be repeated in the same number.

SOLUTION In a five digit number 0 cannot be put in ten thousand's place. So, the number of ways of filling up the ten thousand's place $= 4$.

Since the repetition of digits is allowed, therefore each of the other places can be filled in 5 ways.

So, the required number of numbers $= 4 \times 5 \times 5 \times 5 \times 5 = 2500$.

EXAMPLE 25 How many 4-digit numbers are there, when a digit may be repeated any number of times?

SOLUTION In a four digit number 0 cannot be placed at thousand's place. So, thousand's place can be filled with any digit from 1 to 9. Thus, thousand's place can be filled in 9 ways.

Since repetition of digits is allowed, therefore each of the remaining 3 places can be filled in 10 ways by using the digits from 0 to 9.

Hence, the required number of numbers $= 9 \times 10 \times 10 \times 10 = 9000$.

EXAMPLE 26 How many three-letter words can be formed using a, b, c, d, e if : (i) repetition is not allowed (ii) repetition is allowed?

SOLUTION (i) Clearly, the total number of three-letter words is equal to the number of ways of filling three places. First place can be filled in 5 ways. Now, four letters are left. So, the second place can be filled in 4 ways. Since the repetition of letters is not allowed, so the third place can be filled from any one of the remaining 3 digits in 3 ways.

Hence, total number of words $= 5 \times 4 \times 3 = 60$.

(ii) In this case repetition of letters is allowed, so each of the three places can be filled in 5 ways.

Hence, total number of words $= 5 \times 5 \times 5 = 125$.

EXAMPLE 27 In how many ways can the following prizes be given away to a class of 30 students, first and second in Mathematics, first and second in Physics, first in Chemistry and first in English?

SOLUTION Here we have to give prizes in four subjects and the process of distributing prizes can be completed by giving prizes in the four subjects.

First and second prizes can be given in Mathematics in (30×29) ways.

First and second prizes can be given in Physics in (30×29) ways.

First prize can be given in Chemistry in 30 ways.

First prize can be given in English in 30 ways.

Hence, the number of ways to give prizes in all the four subjects

$$= (30 \times 29) \times (30 \times 29) \times 30 \times 30 = 6.8121 \times 10^8$$

EXAMPLE 28 In how many ways 5 rings of different types can be worn in 4 fingers?

SOLUTION The first ring can be worn in any of the 4 fingers. So, there are 4 ways of wearing it. Similarly, each one of the other rings can be worn in 4 ways.

Hence, the requisite number of ways $= 4 \times 4 \times 4 \times 4 \times 4 = 4^5$.

EXAMPLE 29 In how many ways can 5 letters be posted in 4 letter boxes?

SOLUTION Since each letter can be posted in any one of the four letter boxes. So, a letter can be posted in 4 ways. Since there are 5 letters and each letter can be posted in 4 ways. So, total number of ways in which all the five letters can be posted is $4 \times 4 \times 4 \times 4 \times 4 = 4^5$.

LEVEL-2

EXAMPLE 30 Five persons entered the lift cabin on the ground floor of an 8-floor house. Suppose each of them can leave the cabin independently at any floor beginning with the first. Find the total number of ways in which each of the five persons can leave the cabin (i) at any one of the 7 floors (ii) at different floors.

SOLUTION Suppose A_1, A_2, A_3, A_4, A_5 are five persons.

(i) A_1 can leave the cabin at any of the seven floors. So, A_1 can leave the cabin in 7 ways. Similarly, each of A_2, A_3, A_4, A_5 can leave the cabin in 7 ways. Thus, the total number of ways in which each of the five persons can leave the cabin at any of the seven floors is $7 \times 7 \times 7 \times 7 \times 7 = 7^5$.

(ii) A_1 can leave the cabin at any of the seven floors. So, A_1 can leave the cabin in 7 ways. Now, A_2 can leave the cabin at any of the remaining 6 floors. So, A_2 can leave the cabin in 6 ways. Similarly, A_3, A_4 , and A_5 can leave the cabin in 5, 4 and 3 ways respectively. Thus, the total number of ways in which each of the five persons can leave the cabin at different floors is $7 \times 6 \times 5 \times 4 \times 3 = 2520$.

EXAMPLE 31 A mint prepares metallic calendars specifying months, dates and days in the form of monthly sheets (one plate for each month). How many types of February calendars should it prepare to serve for all the possibilities in the future years?

SOLUTION The mint has to perform two jobs, viz.

- (i) selecting the number of days in the February month (there can be 28 days or 29 days), and
- (ii) selecting the first day of the February month.

The first job can be completed in 2 ways while the second can be performed in 7 ways by selecting any one of the seven days of a week.

Thus, the required number of plates $= 2 \times 7 = 14$.

EXAMPLE 32 For a set of five true/false questions, no student has written all correct answers, and no two students have given the same sequence of answers. What is the maximum number of students in the class, for this to be possible?

SOLUTION Since a true/false type question can be answered in 2 ways either by marking it true or false. So, there are 2 ways of answering each of the 5 questions.

$\therefore$ Total number of different sequences of answers $= 2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$.

Out of these 32 sequences of answers there is only one sequence of answering all the five questions correctly. But no student has written all the correct answers and different students have given different sequences of answers.

$\therefore$ Maximum number of students in the class

$=$ Number of sequences except one sequence in which all answers are correct $= 32 - 1 = 31$

EXAMPLE 33 How many numbers are there between 100 and 1000 such that at least one of their digits is 7?

SOLUTION Clearly, a number between 100 and 1000 has 3-digits

$\therefore$ Total number of 3-digit numbers having at least one of their digits as 7

$=$ (Total number of three-digit numbers) – (Total number of 3-digit numbers in which 7 does not appear at all)

Total number of three-digit numbers: We have to form three-digit numbers by using the digits 0, 1, 2, 3, ..., 9. Clearly, hundred's place can be filled in 9 ways and each of the ten's and one's place can be filled in 10 ways.

So, total number of 3-digit number $= 9 \times 10 \times 10 = 900$.

Total number of three-digit number in which 7 does not appear at all: Here we have to form three-digit numbers by using the digits 0 to 9, except 7. So, hundred's place can be filled in 8

ways and each of the ten's and one's place can be filled in 9 ways. So, total number of three-digit numbers in which 7 does not appear at all is $8 \times 9 \times 9$.

Hence, total number of 3-digit numbers having at least one of their digits as 7 is

$$9 \times 10 \times 10 - 8 \times 9 \times 9 = 252.$$

EXAMPLE 34 How many numbers are there between 100 and 1000 which have exactly one of their digits as 7?

SOLUTION A number between 100 and 1000 contains 3-digits. So, we have to form 3-digit numbers having exactly one of their digits as 7. Such type of numbers can be divided into three types:

- (i) Those numbers that have 7 in the unit's place but not in any other place.
- (ii) Those numbers that have 7 in the ten's place but not in any other place.
- (iii) Those numbers that have 7 in the hundred's place but not in any other place.

Required number of numbers is the total number of these three types of numbers.

We shall now count these three types of numbers separately.

(i) *Those three-digit numbers that have 7 in the unit's place but not in any other place.*

The hundred's place can have any one of the digits from 0 to 9 except 0 and 7. So, hundred's place can be filled in 8 ways. The ten's place can have any one of the digits from 0 to 9 except 7. So, the number of ways the ten's place can be filled is 9. The unit's place has 7. So, it can be filled in only one way.

Thus, there are $8 \times 9 \times 1 = 72$ numbers of the first kind.

(ii) *Those three-digit numbers that have 7 in the ten's place but not in any other place.*

The number of ways to fill the hundred's place = 8

(by any one of the digits from 1, 2, 3, 4, 5, 6, 8, 9)

The number of ways to fill the ten's place = 1 (by 7 only)

The number of ways to fill the one's place = 9 (by any one of the digits 0, 1, 2, 3, 4, 5, 6, 8, 9)

Thus, there are $8 \times 1 \times 9 = 72$ numbers of the second kind.

(iii) *Those three-digit numbers that have 7 in the hundred's place but not at any other place.*

In this case, the hundred's place can be filled only in one way and each of the ten's and one's place can be filled in 9 ways.

So, there are $1 \times 9 \times 9 = 81$ numbers of the third kind.

Hence, the total number of required type of numbers = $72 + 72 + 81 = 225$.

EXAMPLE 35 A telegraph has 5 arms and each arm is capable of 4 distinct positions, including the position of rest. What is the total number of signals that can be made?

SOLUTION Since each arm can be kept in 4 positions and a signal is possible when all the 5 arms are simultaneously placed in positions.

$\therefore$ Total number of ways of placing the arms = $4 \times 4 \times 4 \times 4 \times 4 = 4^5$.

But, this includes one inadmissible case, when all the arms are in the position of rest and then no signal can be made.

Hence, required number of signals = $(4^5 - 1) = 1023$.

EXAMPLE 36 In how many ways can 3 prizes be distributed among 4 boys, when

(i) no boy gets more than one prize? (ii) a boy may get any number of prizes? (iii) no boy gets all the prizes?

SOLUTION (i) The first prize can be given away in 4 ways as it may be given to any one of the 4 boys. The second prize can be given away in 3 ways, because the boy who got the first prize cannot receive the second prize. The third prize can be given away to anyone of the remaining 2 boys in 2 ways. So, the number of ways in which all the prizes can be given away = $4 \times 3 \times 2 = 24$.

ALITER The total number of ways is the number of arrangements of 4 taken 3 at a time. So, the requisite number of ways $= {}^4P_3 = 4! = 24$.

(ii) The first prize can be given away in 4 ways as it may be given to anyone of the 4 boys. The second prize can also be given away in 4 ways, since it may be obtained by the boy who has already received a prize. Similarly, third prize can be given away in 4 ways.

Hence, the number of ways in which all the prizes can be given away $= 4 \times 4 \times 4 = 4^3 = 64$.

(iii) Since any one of the 4 boys may get all the prizes. So, the number of ways in which a boy gets all the 3 prizes is 4.

So, the number of ways in which a boy does not get all the prizes $= 64 - 4 = 60$.

EXAMPLE 37 Find the total number of ways in which n distinct objects can be put into two different boxes.

SOLUTION Let the two boxes be B_1 and B_2 . We observe that there are two choices for each of the n objects. Therefore, by fundamental principle of counting

$$\text{Total number of ways} = \underset{n - \text{times}}{2 \times 2 \times \dots \times 2} = 2^n$$

EXAMPLE 38 Find the total number of ways in which n -distinct objects can be put into two different boxes so that no box remains empty.

SOLUTION Each object can be put either in box B_1 (say) or in box B_2 (say). So, there are two choices for each of the n objects. Therefore, the number of choices for n distinct objects is

$$2 \times 2 \times \dots \times 2 = 2^n. \quad \text{Two of these choices correspond to either the first or the second box being}$$

$n - \text{times}$

empty. Thus, there are $2^n - 2$ ways in which neither box is empty.

EXAMPLE 39 By using the digits 0, 1, 2, 3, 4 and 5 (repetitions not allowed) numbers are formed by using any number of digits. Find the total number of non-zero numbers that can be formed.

SOLUTION Required number of numbers

$$\begin{aligned} &= \text{Number of 1 digit number} + \text{No. of 2 digit numbers} + \dots + \text{Number of 6 digit numbers} \\ &= 5 + 5 \times 5 + 5 \times 5 \times 4 + 5 \times 5 \times 4 \times 3 + 5 \times 5 \times 4 \times 3 \times 2 + 5 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 5 + 25 + 100 + 300 + 600 + 600 = 1630. \end{aligned}$$

EXERCISE 16.2

LEVEL-1

1. In a class there are 27 boys and 14 girls. The teacher wants to select 1 boy and 1 girl to represent the class in a function. In how many ways can the teacher make this selection?
2. A person wants to buy one fountain pen, one ball pen and one pencil from a stationery shop. If there are 10 fountain pen varieties, 12 ball pen varieties and 5 pencil varieties, in how many ways can he select these articles?
3. From Goa to Bombay there are two routes; air, and sea. From Bombay to Delhi there are three routes; air, rail and road. From Goa to Delhi via Bombay, how many kinds of routes are there?
4. A mint prepares metallic calendars specifying months, dates and days in the form of monthly sheets (one plate for each month). How many types of calendars should it prepare to serve for all the possibilities in future years?
5. There are four parcels and five post-offices. In how many different ways can the parcels be sent by registered post?

6. A coin is tossed five times and outcomes are recorded. How many possible outcomes are there?
7. In how many ways can an examinee answer a set of ten true/false type questions?
8. A letter lock consists of three rings each marked with 10 different letters. In how many ways it is possible to make an unsuccessful attempt to open the lock?
9. There are 6 multiple choice questions in an examination. How many sequences of answers are possible, if the first three questions have 4 choices each and the next three have 2 each?
10. There are 5 books on Mathematics and 6 books on Physics in a book shop. In how many ways can a student buy : (i) a Mathematics book and a Physics book (ii) either a Mathematics book or a Physics book?
11. Given 7 flags of different colours, how many different signals can be generated if a signal requires the use of two flags, one below the other? [NCERT]
12. A team consists of 6 boys and 4 girls and other has 5 boys and 3 girls. How many single matches can be arranged between the two teams when a boy plays against a boy and a girl plays against a girl?
13. Twelve students compete in a race. In how many ways first three prizes be given?
14. How many A.P.'s with 10 terms are there whose first term is in the set $\{1, 2, 3\}$ and whose common difference is in the set $\{1, 2, 3, 4, 5\}$?
15. From among the 36 teachers in a college, one principal, one vice-principal and the teacher-incharge are to be appointed. In how many ways can this be done?
16. How many three-digit numbers are there with no digit repeated?
17. How many three-digit numbers are there?
18. How many three-digit odd numbers are there?
19. How many different five-digit number licence plates can be made if
 - (i) first digit cannot be zero and the repetition of digits is not allowed,
 - (ii) the first-digit cannot be zero, but the repetition of digits is allowed?
20. How many four-digit numbers can be formed with the digits 3, 5, 7, 8, 9 which are greater than 7000, if repetition of digits is not allowed?
21. How many four-digit numbers can be formed with the digits 3, 5, 7, 8, 9 which are greater than 8000, if repetition of digits is not allowed?
22. In how many ways can six persons be seated in a row?
23. How many 9-digit numbers of different digits can be formed?
24. How many odd numbers less than 1000 can be formed by using the digits 0, 3, 5, 7 when repetition of digits is not allowed?
25. How many 3-digit numbers are there, with distinct digits, with each digit odd?
26. How many different numbers of six digits each can be formed from the digits 4, 5, 6, 7, 8, 9 when repetition of digits is not allowed?
27. How many different numbers of six digits can be formed from the digits 3, 1, 7, 0, 9, 5 when repetition of digits is not allowed?
28. How many four digit different numbers, greater than 5000 can be formed with the digits 1, 2, 5, 9, 0 when repetition of digits is not allowed?
29. Serial numbers for an item produced in a factory are to be made using two letters followed by four digits (0 to 9). If the letters are to be taken from six letters of English alphabet

without repetition and the digits are also not repeated in a serial number, how many serial numbers are possible?

30. A number lock on a suitcase has 3 wheels each labelled with ten digits 0 to 9. If opening of the lock is a particular sequence of three digits with no repeats, how many such sequences will be possible? Also, find the number of unsuccessful attempts to open the lock.
31. A customer forgets a four-digit code for an Automatic Teller Machine (ATM) in a bank. However, he remembers that this code consists of digits 3, 5, 6 and 9. Find the largest possible number of trials necessary to obtain the correct code.
32. In how many ways can three jobs I, II and III be assigned to three persons A, B and C if one person is assigned only one job and all are capable of doing each job?
33. How many four digit natural numbers not exceeding 4321 can be formed with the digits 1, 2, 3 and 4, if the digits can repeat?
34. How many numbers of six digits can be formed from the digits 0, 1, 3, 5, 7 and 9 when no digit is repeated? How many of them are divisible by 10?
35. If three six faced die each marked with numbers 1 to 6 on six faces, are thrown find the total number of possible outcomes.
36. A coin is tossed three times and the outcomes are recorded. How many possible outcomes are there ? How many possible outcomes if the coin is tossed four times? Five times? n times?
37. How many numbers of four digits can be formed with the digits 1, 2, 3, 4, 5 if the digits can be repeated in the same number?
38. How many three digit numbers can be formed by using the digits 0, 1, 3, 5, 7 while each digit may be repeated any number of times?
39. How many natural numbers less than 1000 can be formed from the digits 0, 1, 2, 3, 4, 5 when a digit may be repeated any number of times?
40. How many five digit telephone numbers can be constructed using the digits 0 to 9. If each number starts with 67 and no digit appears more than once? [NCERT]

LEVEL-2

41. Find the number of ways in which 8 distinct toys can be distributed among 5 children.
42. Find the number of ways in which one can post 5 letters in 7 letter boxes.
43. Three dice are rolled. Find the number of possible outcomes in which at least one die shows 5.
44. Find the total number of ways in which 20 balls can be put into 5 boxes so that first box contains just one ball.
45. In how many ways can 5 different balls be distributed among three boxes?
46. In how many ways can 7 letters be posted in 4 letter boxes?
47. In how many ways can 4 prizes be distributed among 5 students, when
 - (i) no student gets more than one prize?
 - (ii) a student may get any number of prizes?
 - (iii) no student gets all the prizes?
48. There are 10 lamps in a hall. Each one of them can be switched on independently. Find the number of ways in which the hall can be illuminated. [NCERT]

ANSWERS

1. 378 2. 600 3. 6 4. 14 5. 625 6. 32 7. 1024 8. 999
 9. 512 10. (i) 30 (ii) 11 11. 42 12. 42 13. 1320 14. 15 15. 42840
 16. 648 17. 900 18. 450 19. (i) 27216 (ii) 90000 20. 72 21. 48
 22. 720 23. $9(9!)$ 24. 21 25. 60 26. 720 27. 600 28. 48 29. 151200
 30. 720, 719 31. 24 32. 6 33. 229 34. 600, 120 35. 216
 36. 8, 16, 2^n 37. 625 38. 100 39. 215 40. 336 41. 5^8 42. 7^5
 43. 91 44. 20×4^{19} 45. 243 46. 4^7 47. (i) $5!$ (ii) 625 (iii) 620
 48. $2^{10} - 1$.

HINTS TO NCERT & SELECTED PROBLEMS

1. No. of ways = 27×14 .
2. Required number of ways = $10 \times 12 \times 5 = 600$.
3. No of routes = $2 \times 3 = 6$.
4. Total number of calendars = $7 \times 2 = 14$.
5. Since a parcel can be sent to any one of the five post offices. So, required number of ways = $5 \times 5 \times 5 \times 5 = 5^4$.
6. Since toss of each coin can result in 2 ways. So, required no. of ways = $2 \times 2 \times 2 \times 2 \times 2 = 2^5$.
8. Required no. of ways = $10 \times 10 \times 10 - 1$.
9. Each one of the first three questions can be answered in 4 ways and each one of the next three questions can be answered in 2 ways. So, total no. of sequences of answers = $4 \times 4 \times 4 \times 2 \times 2 \times 2$.
11. Required no. of signals = 7×6 .
12. A boy can be selected from the first team in 6 ways, and from the second in 5 ways. So, no. of single matches between the boys of two teams = $6 \times 5 = 30$. Similarly, the no. of single matches between the girls of two teams = $4 \times 3 = 12$. So, total number of matches = $30 + 12 = 42$.
13. Required no. of ways = $12 \times 11 \times 10$.
14. There are 3 ways to choose the first term and corresponding to each such way there are 5 ways of selecting the common difference. So, required no. of A.P.'s = 3×5 .
15. Required no. of ways = $36 \times 35 \times 34$.
16. The total no. of required numbers = $9 \times 9 \times 8$.
17. The total no. of required numbers = $9 \times 10 \times 10$.
18. The total no. of required number = $9 \times 10 \times 5$.
19. (i) Required no. of licence plates = $9 \times 9 \times 8 \times 7 \times 6$
 (ii) Required no. of licence plates = $9 \times 10 \times 10 \times 10 \times 10$.
20. Required no. of numbers = $3 \times 4 \times 3 \times 2$.
21. Required no. of numbers = $2 \times 4 \times 3 \times 2$.
22. Required no. of ways = $6 \times 5 \times 4 \times 3 \times 2 \times 1$.
23. Required no. of numbers = $9 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2$.
24. An odd number less than 1000 may be a one-digit number, two-digit number or a three-digit number. So, required no. of numbers is
 3 (one-digit nos.) + 2×3 (two-digit nos.) + $2 \times 2 \times 3$ (3-digit nos.).
25. Required no. of numbers = $5 \times 4 \times 3$.

26. Required no. of numbers = $6 \times 5 \times 4 \times 3 \times 2 \times 1$.
27. Required no. of numbers = $5 \times 5 \times 4 \times 3 \times 2 \times 1$.
28. Required no. of numbers = $2 \times 4 \times 3 \times 2$.
29. Here we have to perform 6 jobs. So, required number of serial numbers is
 $6 \times 5 \times 10 \times 9 \times 8 \times 7$
30. Required number of sequences = $10 \times 9 \times 8$.
 Also, total number of unsuccessful attempts = $10 \times 9 \times 8 - 1$
31. Number of trials = $4 \times 3 \times 2 \times 1$
32. Required number of ways = $3 \times 2 \times 1$
36. Since a toss of a coin can result in a head or a tail. Therefore, if a coin is tossed n -times, then the total number of outcomes is $2 \times 2 \times 2 \times \dots \times 2 = 2^n$
 $n\text{-times}$
41. Each toy can be distributed in 5 ways.
 So, total number of ways = $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^8$
42. Each letter can be posted in any one of the 7 letter boxes.
 So, required number of ways = $7 \times 7 \times 7 \times 7 \times 7 = 7^5$
43. Required number of possible outcomes
 = Total number of possible outcomes – Number of possible out- comes in which 5 does not appear on any dice.
 $= 6^3 - 5^3 = 216 - 125 = 91$.
44. One ball can be put in first box in 20 ways because we can put any one of the twenty balls in the first box. Now, remaining 19 balls are to be put into remaining 4 boxes. This can be done in 4^{19} ways, because there are 4 choices for each ball. Hence, the required number of ways
 $= 20 \times 4^{19}$.

16.3 PERMUTATIONS

Each of the arrangements which can be made by taking some or all of a number of things is called a permutation.

For example, if there are three objects, then the permutations of these objects, taking two at a time, are

$$ab, \quad ba, \quad bc, \quad cb, \quad ac, \quad ca$$

So, the number of permutations of three different things taken two at a time is 6.

NOTE It should be noted that in permutations the order of arrangement is taken into account; when the order is changed, a different permutation is obtained.

ILLUSTRATION 1 Write down all the permutations of the set of three letters A, B, C.

SOLUTION The permutations of three letters A, B, C taking all at a time are :

$$ABC, \quad ACB, \quad BCA, \quad BAC, \quad CBA, \quad CAB.$$

Clearly, there are 6 permutations.

ILLUSTRATION 2 Write down all the permutations of the vowels A, E, I, O, U in English alphabets taking three at a time, and starting with A.

SOLUTION The permutations of vowels A, E, I, O, U taking three at a time, and starting with A are:

$$AEI, \quad AIE, \quad AEO, \quad AOE, \quad AEU, \quad AUE, \quad AIO, \quad AOI, \quad AIU, \quad AUI, \quad AOU, \quad AUO$$

Clearly, there are 12 permutations.

ILLUSTRATION 3 Write down all the permutations of letters A, B, C, D taking three at a time.

SOLUTION The desired permutations are :

ABC	ABD	BCD	ACD
ACB	ADB	BDC	ADC
BCA	BDA	CBD	CAD
BAC	BAD	CDB	CDA
CAB	DAB	DCB	DAC
CBA	DBA	DBC	DCA

Clearly, there are 24 permutations. These permutations are obtained by first selecting three letters out of 4 and then arranging them in all possible ways.

A NOTATION If n and r are positive integers such that $1 \leq r \leq n$, then the number of all permutations of n distinct things, taken r at a time is denoted by the symbol $P(n, r)$ or ${}^n P_r$.

Thus,

${}^n P_r$ or, $P(n, r)$ = Total number of permutations of n distinct things, taken r at a time.

In illustration 3, we have seen that there are 24 permutations, on a set of 4 letters, taken 3 at a time. Therefore, as per our notation, we have ${}^4 P_3 = 24$ or, $P(4, 3) = 24$.

THEOREM 1 Let r and n be positive integers such that $1 \leq r \leq n$. Then the number of all permutations of n distinct things taken r at a time is given by $n(n-1)(n-2)(n-3) \dots (n-(r-1))$.

i.e. $P(n, r) = {}^n P_r = n(n-1)(n-2) \dots (n-(r-1))$.

PROOF The number of permutations of n distinct things, taken r at a time, is same as the number of ways in which we can fill up r -places when we have n different things at our disposal.

The first place can be filled in n ways, for any one of the n things can be used to fill up the first place. Having filled it, there are $(n-1)$ things left and any one of these $(n-1)$ things can be used to fill up the second place. So, the second place can be filled in $(n-1)$ ways. Hence, by the fundamental principle of counting, the first two places can be filled in $n(n-1)$ ways. When the first two places are filled, there are $(n-2)$ places left, so that the third place can be filled from the remaining $(n-2)$ things in $(n-2)$ ways. Therefore, the first three places can be filled in $n(n-1)(n-2)$ ways. Continuing in this manner, we find that the first $(r-1)$ places can be filled in $n(n-1)(n-2) \dots (n-(r-2))$ ways. After filling up first $(r-1)$ places, exactly $n-(r-1) = n-r+1$ things are left. So, the r th place can be filled in $(n-(r-1))$ ways. Hence, the r places can be filled in $n(n-1)(n-2) \dots (n-(r-1))$ ways.

Hence, the total number of permutations of n distinct things, taken r at a time is

$$n(n-1)(n-2)(n-3) \dots (n-(r-1)).$$

Thus, $P(n, r) = n(n-1)(n-2)(n-3) \dots (n-(r-1))$.

THEOREM 2 Prove that: $P(n, r) = {}^n P_r = \frac{n!}{(n-r)!}$.

PROOF We have,

$$P(n, r) = n(n-1)(n-2)(n-3) \dots (n-(r-1))$$

$$\Rightarrow P(n, r) = \frac{n(n-1)(n-2)(n-3) \dots (n-(r-1))(n-r)(n-(r+1)) \dots 3.2.1}{(n-r)(n-(r+1)) \dots 3.2.1}$$

$$\Rightarrow P(n, r) = \frac{n!}{(n-r)!}$$

THEOREM 3 The number of all permutations of n distinct things, taken all at a time is $n!$.

PROOF The number of all permutations of n distinct things, taken all at a time is same as the number of ways of filling n places when we have n distinct things at our disposal.

Proceeding as in theorem 1, we have

$$P(n, n) = n(n-1)(n-2)(n-3) \dots (n-(n-1)) = n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1 = n!$$

THEOREM 4 Prove that $0! = 1$.

PROOF We have,

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$\Rightarrow P(n, n) = \frac{n!}{0!} \quad [\text{Putting } r = n]$$

$$\Rightarrow n! = \frac{n!}{0!} \quad [\because P(n, n) = n! \text{ (See Theorem 3)}]$$

$$\Rightarrow 0! = \frac{n!}{n!} = 1.$$

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I PROBLEMS BASED UPON THE VALUE OF nP_r OR $P(n, r)$

EXAMPLE 1 Evaluate the following:

(i) 5P_3 (ii) $P(15, 3)$ (iii) $P(5, 5)$

SOLUTION (i) ${}^5P_3 = \frac{5!}{(5-3)!}$

$$\left[\because {}^nP_r = \frac{n!}{(n-r)!} \right]$$

$$\Rightarrow {}^5P_3 = \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2!}{2!} = 60$$

$$(ii) \quad P(15, 3) = \frac{15!}{(15-3)!} = \frac{15!}{12!} = \frac{15 \times 14 \times 13 \times 12!}{12!} = 2730$$

$$(iii) \quad P(5, 5) = \frac{5!}{(5-5)!} = \frac{5!}{0!} = 5! = 120.$$

Type II ON FINDING THE VALUE OF REQUIRED UN-KNOWN WHEN A RELATION CONNECTING $P(n, r)$ IS GIVEN

EXAMPLE 2 If $2 \cdot P(5, 3) = P(n, 4)$, find n .

SOLUTION We have,

$$2 \cdot P(5, 3) = P(n, 4)$$

$$\Rightarrow P(n, 4) = 2 \cdot P(5, 3)$$

$$\Rightarrow \frac{n!}{(n-4)!} = 2 \left(\frac{5!}{(5-3)!} \right)$$

$$\Rightarrow \frac{n(n-1)(n-2)(n-3)(n-4)!}{(n-4)!} = \frac{2(5!)}{2!}$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 5!$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 5 \times 4 \times 3 \times 2 \times 1$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 5(5-1)(5-2)(5-3)$$

$$\Rightarrow n = 5$$

[By comparing two sides]